Lecture #14: Change of Variables

MATH 321 – Mathematical Statistics I

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Finding the Distribution of a Function of Random Variables

We've talked a bit about how to find the distribution of a function of random variables, for certain classes of functions, but let's focus some more on it.

There are a few ways we can try to find the distribution of a function of a random variable (or random variables) whose distribution we know:

1. The Method of Moment-Generating Functions
2. The Method of Distribution Functions
3. The Method of Transformations
4. Using the Jacobian

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The Method of Moment-Generating Functions

There are two important properties of the MGF to remember:

1. If $Y = aX$ where $a \in \mathbb{R}$ is a constant, then:

$$M_Y(t) = M_X(at).$$

2. If $X_1, \ldots, X_n$ are **independent** and $Z = \sum_{i=1}^{n} X_i$, then

$$M_Z(t) = \prod_{i=1}^{n} M_{X_i}(t).$$

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Summary of the Moment-Generating Function Method

Let U be a function of the random variables $Y_1, Y_2, \ldots, Y_n$.

1. Find the moment-generating function for U, $M_U(t)$.
2. Compare $M_U(t)$ with other well-known moment-generating functions. If $M_U(t) = M_V(t)$ for all t in some open interval around 0, then U and V have identical distributions.

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Example: The Method of Moment-Generating Functions

Example: Suppose that $X_1, \ldots, X_n \overset{i.i.d.}{\sim} \text{Exponential}(\beta)$. If $Y = \sum_{i=1}^{n} X_i$, show that the distribution of Y is:

$$Y \sim \text{Gamma}(\alpha = n, \beta = \beta).$$

Example: Let Z be a normally distributed random variable with mean $\mu = 0$ and variance $\sigma^2 = 1$. Use the method of moment-generating functions to find the probability distribution of Z^2.

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Example: A process for refining sugar yields up to 1 ton of pure sugar per day, but the actual amount produced, Y, is a random variable because of machine breakdowns and other slowdowns. Suppose that Y has density function given by

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

The company is paid at the rate of \$300 per ton for the refined sugar, but it also has a fixed overhead cost of \$100 per day. Thus the daily profit, in hundreds of dollars, is $U = 3Y - 1$. Find the probability density function for U.

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Summary of the Distribution Function Method

Let U be a function of the random variables $Y_1, Y_2, \ldots, Y_n$.

1. Find the region $U = u$ in the $(y_1, y_2, \ldots, y_n)$ space.
2. Find the region $U \leq u$.
3. Find $F_U(u) = P(U \leq u)$ by integrating $f(y_1, y_2, \ldots, y_n)$ over the region $U \leq u$.
4. Find the density function $f_U(u)$ by differentiating $F_U(u)$. Thus, $f_U(u) = \dfrac{dF_U(u)}{du}$.

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Example: The Method of Distribution Functions

Example: Let Y have probability density function given by

$$f_Y(y) = \begin{cases} \frac{y+1}{2}, & -1 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the probability density function of $U = Y^2$.

Example: Let U be a uniform random variable on the interval $(0, 1)$. Find a transformation $G(U)$ such that $G(U)$ possesses an exponential distribution with mean β.

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The Method of Transformations

We can take the ideas in that last example further.

Summary of the Transformation Method

Let $U = h(Y)$, where $h(y)$ is a differentiable function that is either strictly increasing or strictly decreasing for all y such that $f_Y(y) > 0$.

1. Find the inverse function, $y = h^{-1}(u)$.
2. Evaluate $\dfrac{dh^{-1}}{du} = \dfrac{d[h^{-1}(u)]}{du}$.
3. Find $f_U(u)$ by

$$f_U(u) = f_Y[h^{-1}(u)] \left| \frac{dh^{-1}}{du} \right|.$$

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Example: The Method of Transformations

Example: Let Y have the probability density function given by:

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Find the density function of $U = -4Y + 3$.

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Example: The Method of Transformations

Example: Consider random variables Y_1, the proportion of impurities in a chemical sample, and Y_2, the proportion of type I impurities among all impurities in the sample. Their joint density function is given by

$$f(y_1, y_2) = \begin{cases} 2(1 - y_1), & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

We are interested in $U = Y_1 Y_2$, which is the proportion of type I impurities in the sample. Find the probability density function for U.

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The Bivariate Transformation Method

Suppose that Y_1 and Y_2 are continuous random variables with joint density function $f_{Y_1,Y_2}(y_1, y_2)$ and that for all (y_1, y_2) such that $f_{Y_1,Y_2}(y_1, y_2) > 0$,

$$u_1 = h_1(y_1, y_2) \qquad \text{and} \qquad u_2 = h_2(y_1, y_2)$$

is a one-to-one transformation from (y_1, y_2) to (u_1, u_2) with inverse

$$y_1 = h_1^{-1}(u_1, u_2) \qquad \text{and} \qquad y_2 = h_2^{-1}(u_1, u_2).$$

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The Bivariate Transformation Method

If $h_1^{-1}(u_1, u_2)$ and $h_2^{-1}(u_1, u_2)$ have continuous partial derivatives with respect to u_1 and u_2 and Jacobian

$$J = \det \begin{bmatrix} \frac{\partial h_1^{-1}}{\partial u_1} & \frac{\partial h_1^{-1}}{\partial u_2} \\ \frac{\partial h_2^{-1}}{\partial u_1} & \frac{\partial h_2^{-1}}{\partial u_2} \end{bmatrix} = \left(\frac{\partial h_1^{-1}}{\partial u_1}\right)\left(\frac{\partial h_2^{-1}}{\partial u_2}\right) - \left(\frac{\partial h_2^{-1}}{\partial u_1}\right)\left(\frac{\partial h_1^{-1}}{\partial u_2}\right) \neq 0,$$

then the joint density of U_1 and U_2 is

$$f_{U_1,U_2}(u_1, u_2) = f_{Y_1,Y_2}\Big(h_1^{-1}(u_1, u_2), h_2^{-1}(u_1, u_2)\Big)|J|,$$

where $|J|$ is the absolute value of the Jacobian, for (u_1, u_2) in the image of the transformation (and 0 elsewhere).

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Example: Multivariable Transformations Using Jacobians

Example: Consider random variables Y_1 and Y_2, with joint density function

$$f(y_1, y_2) = \begin{cases} 2(1 - y_1), & 0 \leq y_1 \leq 1, 0 \leq y_2 \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $U = Y_1 Y_2$, find the probability density function for U by using the bivariate transformation method.

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